

Metric Topology of $\mathbb{C}$

Complex Analysis

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Definition

A complex number z may be thought of as coordinates in the $\mathbb{R} \times i\mathbb{R}$ plane. Such plot is called an **Argand Diagram** and is a common representation of complex numbers.

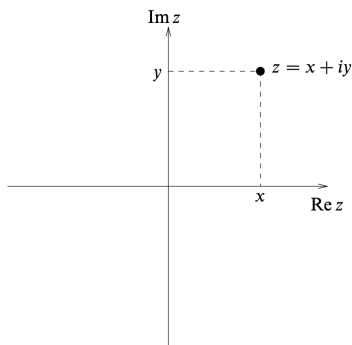


Figure 1: The Argand diagram of a point $z \in \mathbb{C}$

Remark

The addition of complex numbers can also be thought as the sum of coordinates in the $\mathbb{R} \times i\mathbb{R}$ plane.

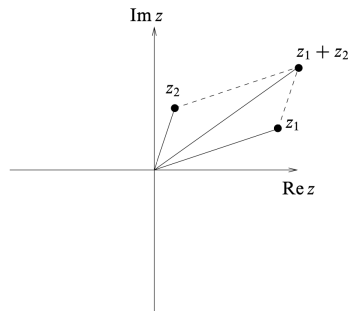


Figure 2: The addition of two complex numbers

Remark

The complex numbers $\pm 1, \pm i$ have modulus unity and using the fact that $|z_1 z_2| = |z_1| |z_2|$ we can see immediately that multiplying a complex number z by one the former, then the latter has the same modulus as z (we identify this operation as a rotation, fact which we will justify later).

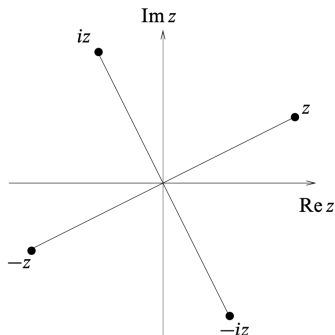


Figure 3: Multiplication of z by $\pm 1, \pm i$

Remark

Complex conjugation corresponds to a reflection of z in the real axis of the Argand diagram.

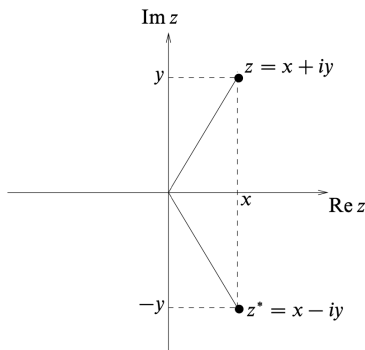


Figure 4: The complex conjugate as a mirror image in the real axis

Definition

A **sequence** of complex numbers (or a **sequence in $\mathbb{C}$**) $Z := (z_n)$ is a function $Z : \mathbb{N} \rightarrow \mathbb{C}$ with domain $\mathbb{N}$ and range in $\mathbb{C}$.

Remark

If $Z : \mathbb{N} \rightarrow \mathbb{C}$ is a sequence, we will usually denote the value of Z at n by the symbol z_n rather than using the function notation $Z(n)$. The values z_n are also called the **terms** or the **elements** of the sequence. We will denote this sequence by the notations

$$Z, \quad (z_n), \quad (z_n : n \in \mathbb{N}).$$

Definition

Let $\varepsilon > 0$, then the **disk of radius ε centred at z_0** is the set

$$D_\varepsilon(z_0) := \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}.$$

A **neighbourhood** of z_0 is any set $V \subseteq \mathbb{C}$ that contains a disk of radius $\varepsilon > 0$ centred at z_0 for some $\varepsilon > 0$. We may also refer to $D_\varepsilon(z_0)$ as the **ε -disk** (or the **ε -ball**) of z_0 .

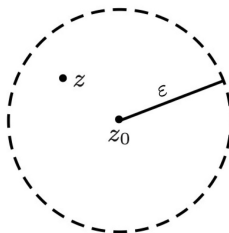


Figure 1: The ε -ball of z_0

Definition

A sequence $Z = (z_n)$ in $\mathbb{C}$ is said to **converge** to $z \in \mathbb{C}$, or z is said to be a **limit** of (z_n) , if for every $\varepsilon > 0$ there exists a natural number K such that $z_n \in D_\varepsilon(z)$ for all $n \geq K$. If a sequence has a limit, we say that the sequence is **convergent**; if it has no limit, we say that the sequence is **divergent**.

Theorem 1.1

Let $Z = (z_n)$ be a sequence of complex numbers, and let $z \in \mathbb{C}$. The following statements are equivalent.

- (a) Z converges to z .
- (b) For every ε , there exists a natural number K such that for all $n \geq K$, the terms z_n satisfy $|z_n - z| < \varepsilon$.
- (c) For every ε -disk $D_\varepsilon(z)$ of z , there exists a natural number K such that for all $n \geq K$, the terms z_n belong to $D_\varepsilon(z)$.

Remark

When a sequence has a limit z , we will use the notation

$$\lim Z = z \quad \text{or} \quad \lim(z_n) = z.$$

We will sometimes use the symbolism $z_n \rightarrow z$, which indicates the intuitive that the values z_n “approach” the number z as $n \rightarrow \infty$.

Definition

A sequence $Z = (z_n)$ of complex numbers is said to be **bounded** if there exists a real number $M > 0$ such that $|z_n| < M$ for all $n \in \mathbb{N}$.

Theorem 1.2

A convergent sequence in $\mathbb{C}$ is bounded.

Theorem 1.3

Let $Z = (z_n)$ and $Z' = (z'_n)$ be sequences in $\mathbb{C}$ that converge to z, z' , respectively, and let $\lambda \in \mathbb{C}$.

- (a) Then the sequences $Z + Z' := (z_n + z'_n)$, $Z - Z' := (z_n - z'_n)$, $ZZ' := (z_n z'_n)$, and $\lambda Z := (\lambda z_n)$ converge to $z + z'$, $z - z'$, zz' , and λz , respectively.
- (b) If $W = (w_n)$ is a sequence of nonzero complex numbers that converges to w and if $w \neq 0$, then the quotient sequence $Z/W := (z_n/w_n)$ converges to z/w .

Theorem 1.4

Let the sequence $Z = (z_n)$ converge to z . Then the sequence $(|z_n|)$ of moduli converges to $|z|$. That is, if $z = \lim(z_n)$, then $\lim(|z_n|) = |z|$.

Theorem 1.5

Let $Z = (z_n)$ be a sequence in $\mathbb{C}$ and let $z \in \mathbb{C}$. If $X = (x_n)$ is a sequence of positive real numbers such that $\lim(x_n) = 0$, and if for some constant $C > 0$ and some $m \in \mathbb{N}$ we have

$$|z_n - z| \leq Cx_n \quad \text{for all } n \geq m,$$

then it follows that $\lim(z_n) = z$.

Theorem 1.6

If $X = (x_n)$ is a sequence in $\mathbb{R}$ and there exist $M \in \mathbb{N}, y \in \mathbb{R}$ such that $x_n \leq y$ for $n \geq M$, then

$$\lim x_n \leq y.$$

Definition

Let $\Omega \subseteq \mathbb{C}$. Then we say that a point $z_0 \in \Omega$ is an **interior point** of Ω if there exists a ball of radius $\rho > 0$ centred at z_0 contained in Ω , that is, $B_\rho(z_0) \subset \Omega$. The **interior** of Ω , denoted by Ω° (or $\text{Int}(\Omega)$) is the set of all interior points of Ω .

Definition

Let $\Omega \subseteq \mathbb{C}$. We say that Ω is **open** if for each point $z_0 \in \Omega$, there exists a ρ -ball $B_\rho(z_0)$ of z_0 such that $B_\rho(z_0) \subset \Omega$, that is, if each point of Ω is an interior point of Ω .

Definition

It is convenient to define the **extended complex plane** $\overline{\mathbb{C}}$ that includes, in addition to $\mathbb{C}$, the **point at infinity** indicated with the symbol ∞ . It plays the role of a complex number in the following sense:

- ▶ $z + \infty = \infty + z = \infty$ for all $z \in \mathbb{C}$.
- ▶ $z\infty = \infty z = \infty$ for all $z \in \mathbb{C}, z \neq 0$.
- ▶ $z/\infty = 0$ for all $z \in \mathbb{C}$.

Definition

Consider in $\mathbb{R}^3$ the sphere with centre $O = (0,0,0)$ and radius 1 and make $\mathbb{C}$ coincide with the plane $x_3 = 0$. The ray coming out from the point $N = (0,0,1)$ and passing through the generic point $P_z = (x,y,0)$ intercepts on the sphere a point Q called the **stereographic projection** of the complex number $z = (x,y)$. The point N is then understood as the stereographic projection of the point at infinity ∞ .

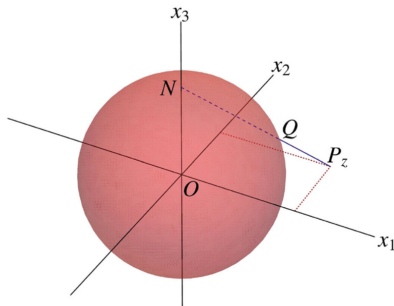


Figure 1: Stereographic projection

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Definition (See the General Topology course for more details)

Let X be a locally compact Hausdorff space. The **one-point compactification of X** is the topological space X^* defined as follows. Let ∞ be some object not in X , and let $X^* = X \sqcup \{\infty\}$ with the following topology:

$$\mathcal{T} = \mathcal{T}_X \cup \{U \subseteq X^* : X^* \setminus U \text{ is a compact subset of } X\}.$$

Then

- i) X^* is a compact Hausdorff space.
- ii) A sequence of points in X diverges to infinity if and only if it converges to ∞ in X^* .
- iii) X is open in X^* and has the subspace topology.

The one-point compactification of $\mathbb{C}$ is called the **Riemann sphere**.

Definition

Let z_1, z_2 be two points in the complex plane. The **line segment ℓ joining z_1 and z_2** is the set

$$\ell := \{z \in \mathbb{C} : z = z_1 + t(z_2 - z_1), 0 \leq t \leq 1\}.$$

Now let $\{z_i\}_{i=1}^{n+1}$ be $n + 1$ points in $\mathbb{C}$. For each $k = 1, 2, \dots, n$ let ℓ_k denote the line segment joining z_k and z_{k+1} . Then the **polygonal path joining z_1 and z_{n+1}** is the set $\bigcup_{i=1}^n \ell_i$. Further, if Ω is a subset of $\mathbb{C}$ and if $\bigcup_{i=1}^n \ell_i \subseteq \Omega$, then we say that the polygonal path joining z_1 and z_{n+1} **lies entirely in Ω** . The polygonal path joining z_1 and z_{n+1} is sometimes written $[z_1, z_2, \dots, z_{n+1}]$ in order to explicitly specify the intermediate points of connection.

Definition

A subset Ω of $\mathbb{C}$ is said to be **connected** if every pair of points z_1, z_2 in Ω can be joined by a polygonal path that lies entirely in Ω . A set that is both open and connected is said to be a **domain** (or a **region**).

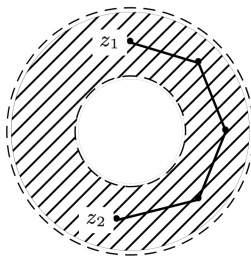


Figure 2: A connected set

Theorem 1.7 (Characterisation of Regions)

A nonempty open subset Ω of $\mathbb{C}$ is connected if and only if it cannot be written as the disjoint union of two nonempty open subsets.

Theorem 1.8

In $\mathbb{C}$, a subset Ω is connected if and only if for each pair of points $z, w \in \Omega$ there is a continuous map $f : [a, b] \rightarrow \Omega$ such that $f(a) = z$ and $f(b) = w$.

Definition

A set $\Omega \subseteq \mathbb{C}$ is said to be **closed** if its complement, denoted by $\Omega^{\sim}$, is open.

Definition

Let $\Omega \subseteq \mathbb{C}$. A point $\omega \in \Omega$ is a **cluster point** (or a **limit point**) of Ω if for every $\rho > 0$ there exists at least one point $z \in \Omega$ such that $0 < |z - \omega| < \rho$. Equivalently, we may say that ω is a cluster point of the set Ω if every ρ -ball $B_\rho(\omega)$ of ω contains at least one point of Ω distinct from ω .

Theorem 1.9

A complex number ω is a cluster point of a subset Ω of $\mathbb{C}$ if and only if there exists a sequence (z_n) in Ω such that $\lim(z_n) = \omega$ and $z_n \neq \omega$ for all $n \in \mathbb{N}$.

Definition

Let $\Omega \subseteq \mathbb{C}$. A point z_0 is said to be an **exterior point** (or an **isolated point**) of Ω if there is some $\rho > 0$ for which the ρ -ball of z_0 not including z_0 is contained in $\Omega^\sim$ or, that is, $B_\rho(z_0) \setminus \{z_0\} \subset \Omega^\sim$ (the ρ -ball of z_0 does not contain any points of Ω distinct from z_0).

Definition

If $\Omega \subseteq \mathbb{C}$ and $z_0 \in \mathbb{C}$, we say that z_0 is a **boundary point** of Ω if every ball centred at z_0 contains at least one point of Ω and one point in $\Omega^{\sim}$.

Thus, a boundary point is neither an interior point nor an exterior point.

The set of all boundary points of Ω is denoted by $\partial\Omega$ and is called the **boundary** (or **frontier**) of Ω .

Definition

The **closure** of $\Omega \subset \mathbb{C}$ is the set

$$\overline{\Omega} := \Omega \cup \partial\Omega.$$

Definition

A set $\Omega \subseteq \mathbb{C}$ is said to be **bounded** if there exists $M > 0$ such that $|z| < M$ whenever $z \in \Omega$, that is, if there exists a ball B of radius M centred at the origin and containing Ω . If Ω is bounded, we define the **diameter** of Ω to be the number

$$\text{diam}(\Omega) = \sup_{z, w \in \Omega} |z - w|.$$

Definition

A subset Ω of $\mathbb{C}$ is said to be **compact** if its it closed and bounded.